

02YMECH - Solution 8

Assigned for the week of Nov 10, 2025

Solutions

1. Let

$$V(r) = -\frac{k}{r}, \quad f(r) = -\frac{k}{r^2},$$

and the orbit be an ellipse with semi-major axis a and eccentricity e . The polar equation is

$$r(\theta) = \frac{a(1 - e^2)}{1 + e \cos \theta}.$$

The angular momentum per unit mass is

$$h = r^2 \dot{\theta} = \sqrt{ka(1 - e^2)}.$$

The time average of any quantity A over one period T is

$$\langle A \rangle = \frac{1}{T} \int_0^T A dt = \frac{1}{T} \int_0^{2\pi} A \frac{r^2}{h} d\theta.$$

Average potential energy

$$\langle V \rangle = -\frac{k}{Th} \int_0^{2\pi} r d\theta.$$

Using the orbit equation,

$$\int_0^{2\pi} r d\theta = a(1 - e^2) \int_0^{2\pi} \frac{d\theta}{1 + e \cos \theta} = 2\pi a \sqrt{1 - e^2}.$$

Thus

$$\langle V \rangle = -\frac{2\pi ka \sqrt{1 - e^2}}{Th}.$$

The period is

$$T = \frac{2\pi a^{3/2}}{\sqrt{k}},$$

so

$$\boxed{\langle V \rangle = -\frac{k}{a}}.$$

Average kinetic energy

The total energy of a Kepler orbit is

$$E = -\frac{k}{2a}.$$

Since $E = \langle T \rangle + \langle V \rangle$,

$$\boxed{\langle T \rangle = \frac{k}{2a}}.$$

Comparison

$$\boxed{\langle T \rangle = -\frac{1}{2}\langle V \rangle},$$

which agrees with the virial relation, obtained here purely from explicit time-average integrals.

2. Two masses m_1, m_2 move in a circular orbit of radius a and period τ . Using the reduced mass μ ,

$$\mu a \omega^2 = \frac{Gm_1 m_2}{a^2}, \quad \omega = \frac{2\pi}{\tau}, \quad \frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}.$$

Hence

$$a = \left(\frac{Gm_1 m_2 \tau^2}{4\pi^2 \mu} \right)^{1/3}.$$

After the motion is stopped, the reduced mass falls radially from rest at $x = a$. Energy conservation gives

$$-\frac{Gm_1 m_2}{a} = \frac{1}{2}\mu \dot{x}^2 - \frac{Gm_1 m_2}{x},$$

so

$$\dot{x} = \sqrt{\frac{2Gm_1 m_2}{\mu} \left(\frac{1}{x} - \frac{1}{a} \right)}.$$

The collision time is

$$t = - \int_a^0 \frac{dx}{\sqrt{\frac{2Gm_1 m_2}{\mu} \left(\frac{1}{x} - \frac{1}{a} \right)}}.$$

Rewrite the integrand:

$$t = - \sqrt{\frac{a\mu}{2Gm_1 m_2}} \int_a^0 \frac{\sqrt{x}}{\sqrt{a-x}} dx.$$

Let $x = y^2$ ($dx = 2y dy$):

$$\int_a^0 \frac{\sqrt{x}}{\sqrt{a-x}} dx = 2 \int_{\sqrt{a}}^0 \frac{y^2}{\sqrt{a-y^2}} dy = -\frac{\pi a}{2}.$$

Therefore

$$t = \frac{\pi a}{2} \sqrt{\frac{\mu}{2Gm_1 m_2}}.$$

Substitute a from above:

$$t = \frac{\pi}{2\sqrt{2}} \sqrt{\frac{a^3}{Gm_1m_2/\mu}} = \frac{\pi}{2\sqrt{2}} \frac{\tau}{2\pi}.$$

$$\boxed{t = \frac{\tau}{4\sqrt{2}}}$$

Thus, after stopping the circular motion, the two particles collide after a time $\tau/(4\sqrt{2})$.

3. Given

$$\lambda(x) = kx^n, \quad 0 \leq x \leq L, \quad n > -1.$$

(a) Total mass:

$$M = \int_0^L kx^n dx = \frac{kL^{n+1}}{n+1}.$$

(b) Center of mass:

$$x_{\text{CM}} = \frac{1}{M} \int_0^L x \lambda(x) dx = \frac{n+1}{n+2} L.$$

(c) Requiring $x_{\text{CM}} = \frac{3L}{4}$:

$$\frac{n+1}{n+2} = \frac{3}{4} \Rightarrow n = 2.$$

For $n = 2$, the mass density increases quadratically with x ; the rod is strongly weighted toward the right end.